A Research on High School Students' Recognition of Using AI-Based Student Assessment Tools

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[Summary]

This research aimed to analyze high school students' recognition of generative AI-based assessment tools by investigating their acceptance, expected effects, and concerns, and to explore the educational utilization potential of AI-based assessment tools based on these findings. A survey was conducted among students from general high schools and specialized high schools. Differences in students' recognition tendencies and between school types were analyzed through reliability analysis, frequency analysis, content analysis, and t-tests. As a result, students generally showed a generally positive acceptance attitude toward utilizing AI as an assessment tool, expressing practical expectations while also voicing concerns about issues such as errors in interpreting answers and the lack of clarity regarding accountability. This result indicates that for AI-based assessment tools to be effectively introduced into educational settings in the future, their design and operation must reflect students' expectations and concerns.

1. Introduction

Recent developments in AI technology are causing various changes in the educational field. In particular, generative AI based on natural language processing, such as ChatGPT, is gaining attention as an innovative tool in the field of educational assessment. The paradigm of educational assessment is shifting from outcome-based evaluation to process-based evaluation. According to the 2022 Revised Curriculum document, the assessment focuses on verifying the extent to which students have achieved educational goals, supplementing areas of learning deficiency, and ultimately improving the quality of teaching and learning. Also, considering the school's circumstances and the characteristics of educational activities, it was recommended to utilize tailored assessments for students through various intelligent information technologies. Seong Tae-je, Si Kija (2024) pointed out, “With the innovative development of digital technology expanding the scope of learning analytics and establishing the foundation for personally tailored assessment, competency-referenced assessments and growth- referenced assessments for supporting the holistic growth of individual students, as well as performance assessments and games grounded in real-world contexts for cultivating future competencies like collaborative problem-solving and creative thinking, are being re-examined, necessitating research on AI-based assessment systems to activate them. [1]”

Accordingly, the study titled “Foundational Research on Utilizing Generative AI as a Student Assessment Tool in Secondary Schools: Based on the Analysis of Secondary Teachers' Perceptions and Experiences with Generative AI” explored the potential and limitations of utilizing generative AI as an assessment tool from the teachers' perspective. The survey results revealed that while teachers held positive expectations, they also expressed concerns about potential adverse effects.

To introduce generative AI-based student assessment in school settings, it is necessary to investigate not only the technical aspects but also users' recognition. Previous studies indicate that middle school teachers show positive intentions toward generative AI-based student assessment. Additionally, middle school teachers anticipate educational benefits from generative AI-based student assessment, including reduced

teacher effort and time, enhanced assessment reliability, provision of tailored feedback, and generation of diverse questions. However, middle school teachers express concerns about dependency and misuse, as well as the provision of biased and inaccurate information [2]. Elementary school teachers also have concerns about misuse and the inaccuracy of information when utilizing ChatGPT in education [3].

These prior discussions delved deeply into teachers' experiences, recognition, expectations, and concerns regarding AI usage from their perspective as the primary users of generative AI-based student assessment tools, thereby illuminating the educational value and potential utilization of AI when employed as an assessment tool. However, it has the limitation of not reflecting the perspective of students who would be the subjects of generative AI-based student assessment tools.

Since AI-based assessment tools are applied directly to students, low acceptance of the tools makes it difficult to expect educational effectiveness, and factors of expectation and concern can be major factors influencing acceptance. Therefore, it is crucial to clearly ascertain how prepared students are to accept AI-based assessment tools, what educational benefits they expect, and what risks or inconveniences they are concerned about. These elements will serve as essential foundational data for establishing an approach method for securing the educational legitimacy of AI-based assessment and its practical introduction. Therefore, this research aims to investigate students' recognition of AI-based student assessment tools from the perspective of the students themselves, who are the parties experiencing educational AI, and to propose directions for an effective and ethical utilization of AI based on its analysis.

Therefore, this research aims to expand upon existing studies targeting middle school teachers and derive an in-depth understanding of AI-based assessment tools, focusing on the recognition of high school students. This research aims to provide practical grounds for the effective introduction and utilization of AI-based student assessment in future school settings by objectively and statistically measuring and analyzing the anticipated acceptance levels, expected effects, and concerns students experience when actually encountering AI assessments. Accordingly, the research questions are set as follows:

1. To what extent are high school students willing to accept generative AI-based assessment tools?
2. How do high school students recognize the expected effects of generative AI-based assessment tools?
3. What concerns do high school students have regarding the utilization of generative AI-based assessment tools?

This paper examines the concept of generative AI-based educational assessment and establishes the overall framework for this research by summarizing the advantages and disadvantages, utilization cases, and ethical issues of AI-based assessment tools based on relevant prior studies. The research method describes in detail the subjects of this research, data collection procedures, survey design methods, and statistical analysis methods, etc. The survey targets students from general high schools and specialized high schools, and the survey content is subject to reliability analysis to verify the consistency of the survey instrument. Following the survey, the results of the respondents' survey are analyzed using frequency analysis, and the content is compared. Additionally, the difference in recognition between students from general high schools and specialized high schools regarding the use of AI-based student assessment tools is visualized through t-tests. Lastly, based on the research results, ways to effectively and ethically utilize AI-based assessment tools in educational settings are proposed, along with a discussion of the research limitations and directions for future follow-up studies.

2. Theoretical Background

2.1 Artificial Intelligence and Generative AI

The High-Level Expert Group on Artificial Intelligence (2019) defined, "Artificial

Intelligence (AI) is a system that demonstrates intelligent behavior by analyzing its environment and taking actions with a degree of autonomy to achieve specific goals" [4]. Since the concept of artificial intelligence was first introduced by Alan Turing and John McCarthy in the 20th century, AI has continued to evolve and is now firmly established in our daily lives. Specifically, Sengar, S.S., Hasan, A.B., and Kumar, S. (2024) suggested, "Generative AI is a type of artificial intelligence that refers to AI systems capable of creating text, images, or other types of media using generative models" [5]. Additionally, they stated, "These models gain an understanding of patterns and structures within the learning data and subsequently generate new data with similar characteristics." This research empirically explores high school students' willingness to accept, anticipated educational effects, and potential concerns regarding an advanced assessment method that utilizes generative AI to grade students' essay-type responses in real time, precisely analyze individual learning characteristics and error patterns, and provide tailored feedback.

2.2 Artificial Intelligence (AI) and Education

Discussions on utilizing artificial intelligence in education are actively underway, intertwined with shifts in educational paradigms and the commercialization of AI, etc. Since 2017, China has published more than four AI textbooks for use in schools, and Finland has also actively incorporated AI into education, such as developing the program 'Elements of AI', a six-week AI learning program, in 2018 through a collaboration between the Finnish government, the University of Helsinki, and the Finnish company Reaktor [6].

Amidst this trend, artificial intelligence is gaining attention not only as a tool for innovating teaching methods but also for complementing existing assessment methods. Examples include C-Rater-ML, developed by the Educational Testing Service (ETS) for diagnosing math and science concepts, automated scoring research on acid-base concepts, and automated scoring research on photosynthesis concepts. However, there are concerns regarding AI-based scoring, including negative recognition of AI, errors or accuracy issues, adverse effects from incorrect predictions, and technical limitations in

identifying cultural meanings embedded within sentences [7]. Therefore, to effectively introduce AI into educational assessment settings, research on user recognition and acceptance attitudes is essential, not merely the technological excellence of the system.

2.2.1 Essay-type Assessment Utilizing AI: WA3I Case

Ha, Minsu et al. (2019) developed WA3I, an AI-based automatic scoring program for essay-type questions [8]. In this research, an automatic scoring model was constructed for questions with high agreement between computer and expert scoring by developing essay-type assessment questions and collecting student responses. The pilot operation results showed that WA3I demonstrated scoring results similar to those of experts in the automatic scoring of essay-type questions and was able to provide immediate feedback. From the teacher's perspective, it is possible to expect that scoring time can be reduced, consistency in evaluation can be ensured, and objective assessments can be conducted. To enhance the utilization to a level that is reliable in the field, the accuracy and convenience of the program must be improved, and since the accuracy of such programs is based on student responses, systematic data management is required, suggesting the need for close collaboration between the field and research.

2.3 Students' Recognition of Learning Environment and Learning Approach

In the Presage stage of the 3P model presented by Biggs (2011), “Perception of the learning environment” was identified as a key variable, with the paper asserting that "Students' perceptions of the learning context significantly influence their approaches to learning, which in turn lead to differences in the quality of learning outcomes."

Furthermore, one longitudinal study noted, “There are significant positive concurrent correlations between learning environment perception, …, and deep learning approach among students” [10]. This suggests that the more students perceive their learning environment as useful, the higher their goal orientation becomes, leading to more effective learning and significantly impacting actual learning outcomes.

3. Research Method

3.1 Research Subject

This research was conducted on 83 students from general high schools and 76 students from specialized high schools.

3.2 Research Tool

This research analyzes high school students' acceptance, expectations, and concerns regarding generative AI-based assessment and examines their recognition to effectively utilize it in educational settings. The survey employed in this research was designed to analyze high school students' acceptance, expected effects, and concerns regarding generative AI-based student assessment. The survey questions were based on the survey questions designed for teachers in the study by Shin Sangyun and Kang Shincheon (2024). The survey questions consisted of 5 questions on acceptance, 5 questions on expected effects, 5 questions on concerns, and 2 open-ended questions [2]. All multiple-choice questions were structured to be answered on a 5-point Likert scale. Subsequent reliability analysis of the questions using SPSS yielded a Cronbach's α value of .737, indicating high reliability of the responses. This demonstrates that the questions within each domain possess reliable internal consistency. The following summarizes the structure and key content of the survey questions actually used.

Acceptance
I have experience using generative AI-based tools for learning or assignments.
I have experience using generative AI in my daily life.
I think the results scored by generative AI can be trusted.
I think it's okay for me to be evaluated utilizing generative AI-based assessment tools.
I think using generative AI-based assessment tools is necessary for future education.

Expected Effects

I think generative AI-based assessment tools will have a positive impact on future school education.
I think generative AI-based assessments can detect cheating.
I think generative AI-based assessments can objectively diagnose my learning level.
I think generative AI assessment tools could reduce teachers' evaluation burdens and improve the quality of teaching.
I think generative AI will provide me with more accurate feedback.

Concerns (reverse-worded questions, unlike acceptance and expected effects)
I think generative AI-based assessments might misinterpret my answers.
I think generative AI assessment tools pose a risk of personal information leakage.
I think the results assessed by generative AI may be unfair or biased.
I think assessments through generative AI may lead to unclear accountability for the assessment results.
I think generative AI may not properly understand my essay-type answers.

3.3 Data Collection

An internet survey was conducted using Google Forms from May 23 to May 30, 2025. The general high school students who responded to the survey were 11 first grade male students, 12 first grade female students, 27 second grade male students, and 33 second grade female students. The survey was conducted at a specialized high school with 62 first grade male students and 14 first grade female students. Before the survey, the concept and features of AI-based assessment were explained to students during class time and breaks between classes, and responses were collected anonymously. The independent variable in this research is school type (general or specialized), and the dependent variables are acceptance, expected effects, and concerns.

<Table 3-1> Participants

General high school	First grade student	Second grade student

student		
Male student	11 people	27 people
Female student	12 people	33 people

Specialized high school student	First grade student	Second grade student
Male student	62 people	0 person
Female student	14 people	0 person

3.4 Data Analysis

First, exploratory factor analysis will be conducted to verify the validity and reliability of the survey questions. In this process, only questions with factor loading of 0.4 or higher will be adopted. Additionally, Cronbach's α coefficient will be calculated to verify the internal consistency of the survey questions ($\alpha \geq 0.7$).

Second, Pearson Correlation Analysis will be conducted to identify the relationship between ‘expectations’, ‘concerns’, and ‘willingness to accept’ for question analysis. Subsequently, Multiple Regression Analysis will be conducted to identify factors influencing willingness to accept.

Third, this research conducts frequency analysis using SPSS Statistics 26.0 based on the collected survey response data to identify respondents' recognition trends by gender and grade level. This identifies the overall distribution of students' recognition regarding generative AI-based assessment tools and compares response trends.

Lastly, to analyze the difference in recognition between students from general high schools and specialized high schools, independent samples t-tests and one-way analysis of variance (ANOVA) are conducted using Python's scipy and pandas libraries. All analysis results are visually presented utilizing visualization tools.

For open-ended subjective questions, the collected subjective responses will be lightly reviewed, and similar expressions will be categorized by semantic units into predefined keywords. Subsequently, Python's automatic classification algorithm will be utilized to analyze the frequency of recognition trends within the categorized data.

4. Research Result

4.1 Research Question

This research sets the following research questions to explore the acceptance, expected effects, and concerns that high school students actually hold regarding the assessment of generative AI.

1. To what extent are high school students willing to accept generative AI-based assessment tools?
2. How do high school students recognize the expected effects of generative AI-based assessment tools?
3. What concerns do high school students have regarding the utilization of generative AI-based assessment tools?

To verify the reliability of the multiple-choice questions in the survey, Cronbach's α coefficient was calculated. The reliability was .737 (α > 0.7), indicating a high level of reliability of the responses.

Reliability Statistics

Cronbach's Alpha	N of Items
.737	15

[Table 4-1] Validation of the reliability of multiple-choice survey questions using Cronbach's α

The mean and standard deviation of responses to multiple-choice questions, excluding open-ended questions, in the survey are as shown in [Table 4-2]. Among the multiple-choice questions, questions 1 and 5, which were about the respondents' experience using generative AI, had the highest mean scores (M = 4.62, M = 4.64). In other words, high school students were found to have extensive experience using generative AI both in their learning and assignments as well as in their daily lives. Meanwhile, questions 2 and 14 showed the lowest mean scores (M = 3.29, M = 3.21). On the other hand, questions 3 and

14 exhibited the highest standard deviations among all questions, at 1.159 and 1.131 respectively.

Item Statistics

		Mean	Std. Deviation	N
1.	I have experience using generative AI-based tools for learning or assignments.	4.62	.654	159
2	I think the results scored by generative AI can be trusted.	3.29	.859	159
3	I think it's okay for me to be evaluated utilizing generative AI-based assessment tools.	3.35	1.159	159
4	I think using generative AI-based assessment tools is necessary for future education.	3.99	.958	159
5	I have experience using generative AI in my daily life.	4.64	.566	159
6	I think generative AI-based assessment tools will have a positive impact on future school education.	3.84	.954	159
7	I think generative AI-based assessments can detect cheating.	3.63	.991	159
8	I think generative AI-based assessments can objectively diagnose my learning level.	3.48	1.036	159
9	I think generative AI assessment tools could reduce teachers' evaluation burdens and improve the quality of teaching.	3.84	.945	159
10	I think generative AI will provide me with more accurate feedback.	3.67	.978	159
12.	I think generative AI-based assessments might misinterpret my answers.	4.06	.929	159
13.	I think generative AI assessment tools pose a risk of personal information leakage.	3.86	.913	159
14.	I think the results assessed by generative AI may be unfair or biased.	3.21	1.131	159
15.	I think assessments through generative AI may lead to unclear accountability for the assessment results.	4.00	.921	159
16.	I think generative AI may not properly understand my essay-type answers.	3.84	.965	159

[Table 4-2] Mean, standard deviation, and number of responses for multiple-choice survey questions

The multiple-choice questions in the survey were categorized into three factors: acceptance, expected effects, and concerns, and descriptive statistical analysis was conducted. The results are shown in [Table 4-3]. The mean for questions 1 to 5 related to acceptance was 3.9786, with a standard deviation of .56729. The mean for questions 6 to 10 related to expected effects was 3.6918, with a standard deviation of .71609. Lastly, the mean for questions 12–16 related to concerns was 3.7925, with a standard deviation of .66481.

Descriptive Statistics

	N	Mean	Std. Deviation
Acceptance	159	3.9786	.56729
Expected effects	159	3.6918	.71609
Concerns	159	3.7925	.66481
Valid N (listwise)	159		

[Table 4-3] Mean and standard deviation by survey factor

Next, Pearson correlation analysis was conducted to analyze the correlations among acceptance, expected effects, and concerns. The results are shown in [Table 4-4]. No significant correlation was found between acceptance and concerns ($r = -.022$, $p = .784$) and between expected effects and concerns ($r = -.068$, $p = .393$). On the other hand, acceptance and expected effects were found to have a statistically significant positive correlation ($r = .627$, $p < .001$).

Correlations

		Acceptance	Expected effects	Concerns
Acceptance	Pearson Correlation	1	.627**	-.022
	Sig. (2-tailed)		<.001	.784
	N	159	159	159
Expected effects	Pearson Correlation	.627**	1	-.068
	Sig. (2-tailed)	<.001		.393
	N	159	159	159
Concerns	Pearson Correlation	-.022	-.068	1
	Sig. (2-tailed)	.784	.393	
	N	159	159	159

**. Correlation is significant at the 0.01 level (2-tailed).

[Table 4-4] Correlation analysis between acceptance, expected effects, and concerns

4.2 **Frequency Analysis** Result

4.2.1 Acceptance of Generative AI Assessment Tools

The results of analyzing high school students' responses regarding their experiences using generative AI are shown in [Table 4-5] below.

		Frequency	Percentage	Valid percentage	Cumulative percentage
Valid	2	1	.6	.6	.6
	3	4	2.4	2.5	3.1
	4	46	27.1	28.7	31.9
	5	109	64.1	68.1	100.0
	Total	160	94.1	100.0	
Missing	system	10	5.9		
Total		170	100.0		

[Table 4-5] Distribution of responses regarding experiences using generative AI in daily life

Of the 160 valid respondents, 109 (68.1%) answered, “Strongly agree” (5 points), while 46 (28.7%) answered, “Agree” (4 points). Meanwhile, 4 respondents (2.5%) rated it as ‘Average’ (3 points), 1 respondent (0.6%) rated it as ‘Disagree’ (2 points), and none of the students had no experience using it at all. Overall, approximately 96.8% of respondents responded that they had experience using generative AI in their daily lives.

Meanwhile, the results of confirming experiences using generative AI tools for learning or assignments are shown in the table below.

		Frequency	Percentage	Valid percentage	Cumulative percentage
Valid	2	3	1.8	1.9	1.9
	3	6	3.5	3.8	5.6
	4	40	23.5	25.0	30.6
	5	111	65.3	69.4	100.0
	Total	160	94.1	100.0	

[Table 4-6] Distribution of responses regarding experience using generative AI tools for learning/assignments

Of the 160 valid respondents, 111 (69.4%) answered, “Strongly agree” (5 points), and 40 (25.0%) answered, “Agree” (4 points). ‘Average’ (3 points) was selected by 6 respondents (3.8%), and ‘Disagree’ (2 points) by 3 respondents (1.9%). No students reported having no experience using such tools. Approximately 94.4% of respondents responded that they had actually used AI tools for learning or assignments.

The analysis results regarding whether the results scored by generative AI can be trusted are shown in [Table 4-7].

		Frequency	Percentage	Valid percentage	Cumulative percentage
Valid	1	4	2.4	2.5	2.5
	2	25	14.7	15.6	18.1
	3	62	36.5	38.8	56.9
	4	61	35.9	38.1	95.0
	5	8	4.7	5.0	100.0
	Total	160	94.1	100.0	
Missing	system	10	5.9		
Total		170	100.0		

[Table 4-7] Distribution of reliability recognition of the result of generative AI scoring

Of the 160 respondents, 62 (38.8%) answered, “Average” (3 points), 61 (38.1%) answered ‘'Agree” (4 points), and 8 (5.0%) answered, “Strongly agree” (5 points). Meanwhile, 25 respondents (15.6%) answered, “Disagree” (2 points), and 4 respondents (2.5%) answered, ‘“Strongly disagree” (1 point). Overall, approximately 81.9% responded that they trust AI scoring results at least to the extent of ‘Average’.

The results of the recognition regarding the willingness to accept evaluation using generative AI assessment tools are shown in Table [4-8].

		Frequency	Percentage	Valid percentage	Cumulative percentage

		빈도	퍼센트	유효 퍼센트	누적 퍼센트
Valid	1	13	7.6	8.1	8.1
	2	27	15.9	16.9	25.0
	3	38	22.4	23.8	48.8
	4	57	33.5	35.6	84.4
	5	25	14.7	15.6	100.0
	Total	160	94.1	100.0	
Missing	system	10	5.9		
Total		170	100.0		

[Table 4-8] Distribution of recognition of willingness to accept evaluation using generative AI assessment tools

Of the 160 respondents, 38 (23.8%) rated it 'Average' (3 points), 57 (35.6%) rated it 'Agree' (4 points), and 25 (15.6%) rated it 'Strongly agree' (5 points). Conversely, 27 respondents (16.9%) rated it as 'Disagree' (2 points), and 13 respondents (8.1%) rated it as 'Strongly disagree' (1 point).

The analysis of recognition regarding the future educational necessity of utilizing generative AI assessment tools is shown in Table [4-9].

		Frequency	Percentage	Valid percentage	Cumulative percentage
Valid	1	13	7.6	8.1	8.1
	2	27	15.9	16.9	25.0
	3	38	22.4	23.8	48.8
	4	57	33.5	35.6	84.4
	5	25	14.7	15.6	100.0
	Total	160	94.1	100.0	
Missing	system	10	5.9		
Total		170	100.0		

[Table 4-9] Distribution of recognition of necessity to introduce generative AI assessment tools

Of those respondents, 38 (23.8%) answered, "Average" (3 points), 57 (35.6%) answered, ''Agree" (4 points), and 25 (15.6%) answered, "Strongly agree" (5 points). Twenty-seven respondents (16.9%) answered, "Disagree" (2 points), and 13 respondents (8.1%) answered, "Strongly disagree" (1 point). Overall, 75% of respondents showed a positive recognition of at least average.

4.2.2 Expected Effects of Generative AI Assessment Tool

The analysis results regarding high school students' recognition of the positive educational impact of generative AI assessment tools are shown in [Table 4-10].

		Frequency	Percentage	Valid percentage	Cumulative percentage
Valid	1	3	1.8	1.9	1.9
	2	12	7.1	7.5	9.4
	3	33	19.4	20.6	30.0
	4	71	41.8	44.4	74.4
	5	41	24.1	25.6	100.0
	Total	160	94.1	100.0	
Missing	system	10	5.9		
Total		170	100.0		

[Table 4-10] Distribution of recognition of positive educational impacts of generative AI assessment tools

Of the 160 valid respondents, ‘Agree’ (4 points) was the most common response at 71 respondents (44.4%), followed by ‘Strongly agree’ (5 points) at 41 respondents (25.6%). ‘Average’ (3 points) was selected by 33 respondents (20.6%), ‘Disagree’ (2 points) by 12 respondents (7.5%), and ‘Strongly disagree’ (1 point) by 3 respondents (1.9%). Overall, 70.0% of all respondents responded that AI assessment tools will have a positive impact on future school education.

The analysis results regarding recognition of expectations for the potential of generative AI-based assessment to detect cheating are summarized in [Table 4-11].

		Frequency	Percentage	Valid percentage	Cumulative percentage
Valid	1	4	2.4	2.5	2.5
	2	18	10.6	11.3	13.8
	3	41	24.1	25.6	39.4
	4	68	40.0	42.5	81.9
	5	29	17.1	18.1	100.0
	전체	160	94.1	100.0	
결측	Total	10	5.9		
Missing	system	170	100.0		
Total					

[Table 4-11] Distribution of recognition of the potential of detecting cheating in generative AI-based assessments

68 respondents (42.5%) answered, “Agree” (4 points) and 29 respondents (18.1%) answered, ''Strongly agree” (5 points), meaning 60.6% of respondents believed AI evaluation can detect cheating. Meanwhile, 41 respondents (25.6%) rated it ‘Average’ (3 points), 18 respondents (11.3%) rated it ‘Disagree’ (2 points), and 4 respondents (2.5%) rated it ‘Strongly disagree’ (1 point).

The results confirming their recognition of expectations regarding the potential of generative AI-based assessment to diagnose learning levels are shown in [Table 4-12].

		Frequency	Percentage	Valid percentage	Cumulative percentage
Valid					
	Total				
Missing	system				
Total					

[Tabel 4-12] Distribution of recognition of the potential for generative AI-based assessment to diagnose learning levels

'Agree' (4 points) was selected by 62 respondents (38.8%), 'Strongly agree' (5 points) by 25 respondents (15.6%), and 'Average' (3 points) by 40 respondents (25.0%). Meanwhile, negative responses were 'Disagree' (2 points) for 29 people (18.1%) and 'Strongly disagree' (1 point) for 4 people (2.5%). Responses of 'Average' or higher accounted for 73.8% of the total.

The results confirming recognition of expectations regarding AI assessment tools' reduction of teachers' workload and improvement in lesson quality are shown in [Table 4-13].

		Frequency	Percentage	Valid percentage	Cumulative percentage
Valid					

		빈도	퍼센트	유효 퍼센트	누적 퍼센트
유효	1	4	2.4	2.5	2.5
		29	17.1	18.1	20.6
		40	23.5	25.0	45.6
	4	62	36.5	38.8	84.4
	5	25	14.7	15.6	100.0
	전체	160	94.1	100.0	
결측	시스템	10	5.9		
전체		170	100.0		

		빈도	퍼센트	유효 퍼센트	누적 퍼센트
유효	1	4	2.4	2.5	2.5
	2	11	6.5	6.9	9.4
	3	31	18.2	19.4	28.7
	4	76	44.7	47.5	76.3
	5	38	22.4	23.8	100.0
	전체	160	94.1	100.0	
결측	시스템	10	5.9		
전체		170	100.0		

[Table 4-13] Distribution of recognition of AI assessment tools' reduction of teachers' workload and improvement in lesson quality

Of those who responded, 76 respondents (47.5%) answered, “Agree” (4 points), and 38 respondents (23.8%) answered, “Strongly agree” (5 points), meaning approximately 71.3% of all respondents showed a positive recognition. ‘Average’ (3 points) was selected by 31 respondents (19.4%), while ‘Disagree’ (2 points) and ‘Strongly disagree’ (1 point) were chosen by 11 (6.9%) and 4 (2.5%) respondents, respectively.

The analysis of recognition of expectations regarding generative AI's ability to provide accurate feedback is shown in [Table 4-14].

		Frequency	Percentage	Valid percentage	Cumulative percentage
Valid					
Total					
Missing	system				
Total					

[Table 4-14] Distribution of recognition of generative AI's ability to provide accurate feedback

‘Agree’ (4 points) was selected by 68 respondents (42.5%), and ‘Strongly agree’ (5 points) by 31 respondents (19.4%), meaning 61.9% of the total positively recognized the accuracy of AI feedback. ‘Average’ (3 points) was chosen by 41 respondents (25.6%), ‘Disagree’ (2 points) by 16 respondents (10.0%), and ‘Strongly disagree’ (1 point) by 4 respondents (2.5%).

		빈도	퍼센트	유효 퍼센트	누적 퍼센트
유효	1	4	2.4	2.5	2.5
	2	16	9.4	10.0	12.5
	3	41	24.1	25.6	38.1
	4	68	40.0	42.5	80.6
	5	31	18.2	19.4	100.0
	전체	160	94.1	100.0	
결측	시스템	10	5.9		
전체		170	100.0		

4.2.3 Concerns Regarding Generative AI Assessment Tools

The analysis of concerns regarding the potential for errors in interpreting responses in generative AI assessments is shown in [Table 4-15].

	Frequency	Percentage	Valid percentage	Cumulative percentage
Valid				

		빈도	퍼센트	유효 퍼센트	누적 퍼센트
유효	1	1	.6	.6	.6
	2	13	7.6	8.2	8.9
	3	18	10.6	11.4	20.3
	4	68	40.0	43.0	63.3
	5	58	34.1	36.7	100.0
	전체	158	92.9	100.0	
결측	시스템	12	7.1		
전체		170	100.0		

[Table 4-15] Distribution of recognition of potential errors in interpreting responses during generative AI assessments

Of the 158 valid respondents, 68 (43.0%) answered, “Agree” (4 points) and 58 (36.7%) answered, “Strongly agree” (5 points), meaning 79.7% of the total expressed concern about the potential for interpretation errors. ‘Average’ (3 points) was selected by 18 respondents (11.4%), ‘Disagree’ (2 points) by 13 respondents (8.2%), and ‘Strongly disagree’ (1 point) by 1 respondent (0.6%).

Next, the results regarding recognition of concerns regarding the potential leakage of personal information from generative AI assessment tools are shown in [Table 4-16].

		빈도	퍼센트	유효 퍼센트	누적 퍼센트
유효	1	2	1.2	1.3	1.3
	2	10	5.9	6.3	7.6
	3	37	21.8	23.4	31.0
	4	70	41.2	44.3	75.3
	5	39	22.9	24.7	100.0
	전체	158	92.9	100.0	
결측	시스템	12	7.1		
전체		170	100.0		

		Frequency	Percentage	Valid percentage	Cumulative percentage
Valid					
Missing	system				
Total					

[Table 4-16] Distribution of recognition of potential personal information leakage in generative AI assessments

Of the 158 respondents, 70 (44.3%) answered, “Agree” (4 points) and 39 (24.7%) answered, “Strongly agree” (5 points), meaning approximately 69.0% of the total expressed concern about the potential for personal information leaks. ‘Average’ (3 points) was selected by 37 people (23.4%), ‘Disagree’ (2 points) by 10 people (6.3%), and ‘Strongly disagree’ (1 point) by 2 people (1.3%).

The analysis of recognition of concerns regarding potential bias and unfairness in generative AI assessment results is shown in [Table 4-17].

		Frequency	Percentage	Valid percentage	Cumulative percentage
Valid	1	12	7.1	7.6	7.6
	2	34	20.0	21.5	29.1
	3	40	23.5	25.3	54.4
	4	53	31.2	33.5	88.0
	5	19	11.2	12.0	100.0
	Total	158	92.9	100.0	
Missing	system	12	7.1		
Total		170	100.0		

[Table 4-17] Distribution of recognition of potential bias and unfairness in generative AI assessments

Of the 158 respondents, 53 (33.5%) answered, “Agree” (4 points), 19 (12.0%) answered, “Strongly agree” (5 points), and 40 (25.3%) answered, “Average” (3 points). ‘Disagree’ (2 points) was selected by 34 respondents (21.5%), and ‘Strongly disagree’ (1 point) by 12 respondents (7.6%). Overall, 71.1% of respondents answered at an average level or above.

The analysis of recognition regarding of concerns for the unclear accountability generative results of is shown in AI assessments [Table 4-18].

		빈도	퍼센트	유효 퍼센트	누적 퍼센트
유효	1	1	.6	.6	.6
	2	11	6.5	7.0	7.6
	3	28	16.5	17.7	25.3
	4	65	38.2	41.1	66.5
	5	53	31.2	33.5	100.0
	전체	158	92.9	100.0	
결측	시스템	12	7.1		
전체		170	100.0		

		빈도	퍼센트	유효 퍼센트	누적 퍼센트
유효	1	1	.6	.6	.6
	2	1	.6	0.8	1.4
	3	30	17.6	19.0	30.4
	4	66	38.8	41.8	72.2
	5	44	25.9	27.8	100.0
	전체	158	92.9	100.0	
결측	시스템	12	7.1		
전체		170	100.0		

[Table 4-18] Distribution of recognition of potential bias and unfairness in generative AI assessments

Of the 158 respondents, 65 (41.1%) answered, “Agree” (4 points) and 53 (33.5%) answered, “Strongly agree” (5 points), indicating that the majority expressed concern. Twenty-eight respondents (17.7%) rated it as ‘Average’ (3 points), while only 12 respondents (7.6%) gave negative responses. Overall, 74.6% expressed at least average concern about the lack of clarity regarding accountability.

Finally, the results regarding the recognition of concerns about generative AI's potential lack of understanding in essay-typ

[Table 4-19] Distribution of recognition of potential lack of understanding for essay-type responses in generative AI

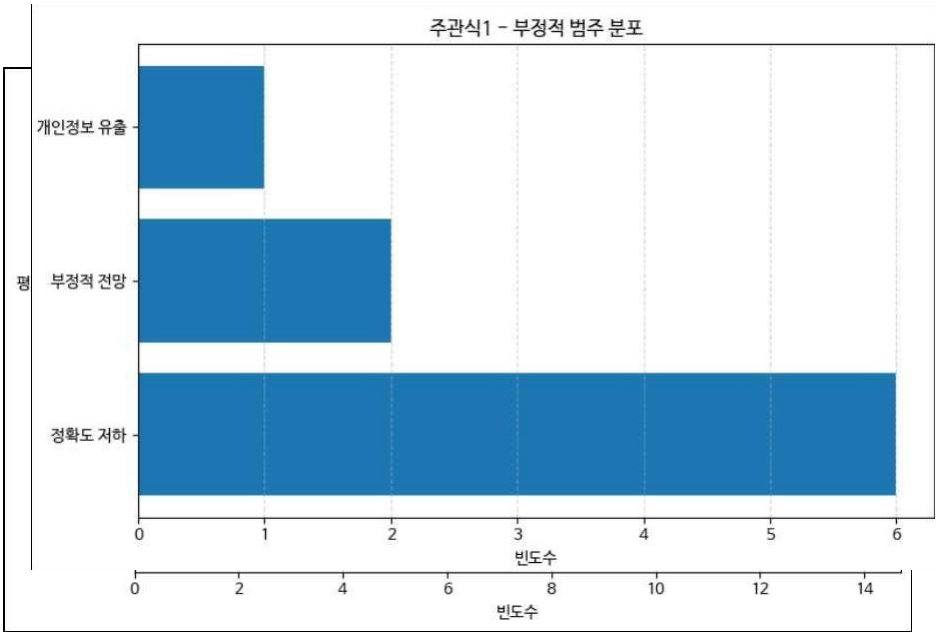
66 respondents (41.8%) answered, “Agree” (4 points), while 44 respondents (27.8%) answered, “Strongly agree” (5 points), indicating 69.6% of all respondents recognized that AI may not accurately understand essay-type responses. ‘Average’ (3 points) was selected

by 30 respondents (19.0%), while negative responses accounted for only 11.4%.

Questions 11 and 17 were open-ended subjective questions, prompting students to freely describe their positive expectations and negative concerns regarding generative AI-based assessment tools. These responses were organized by categorizing similar expressions by semantic units, where each category was classified based on predefined keywords.

Categorization was performed using Python-based automatic classification algorithms, resulting in ten major categories, including ‘Increased Fairness’, ‘Improved Assessment Accuracy’, ‘Personal Information Leakage’, and ‘Social Issues’. Subsequently, the frequency of each category was calculated to identify students' recognition trends.

Responses to Question 11 (“If you have any expectations regarding generative AI-based assessment tools, please feel free to write them down.”) were categorized as follows, being centered on positive recognition. The most frequent category was ‘Positive Outlook’ (14 instances), with repeated expressions such as “It will continue to advance” and “It seems like it will teach me things I don't know.” Additional mentions included ‘‘Faster Speed’' (12 instances), ‘‘Increased Assessment Accuracy’' (9 instances), and



‘‘Enhanced Fairness’' (4 instances).

[Figure 4-20] Positive ex

Essay-type 1 – Positive Category Distribution

On the

Enhanced Fairness

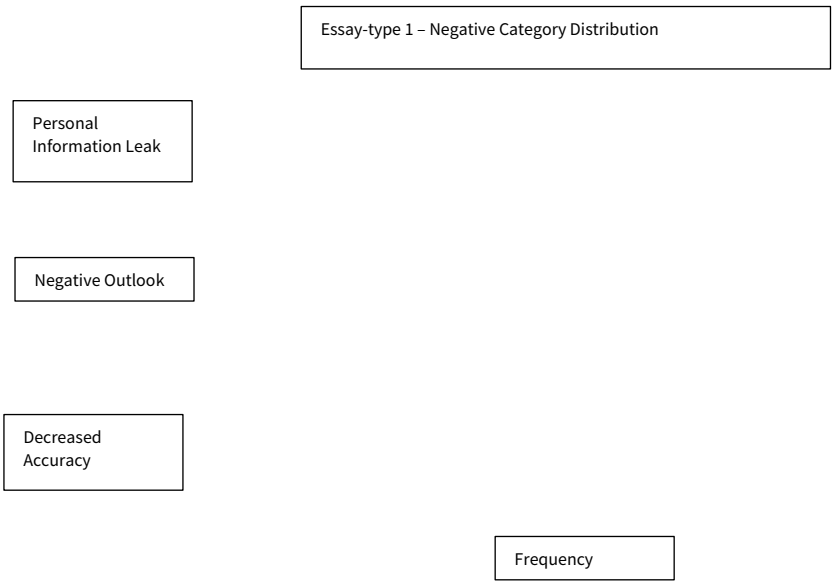
d, some responses expressed negative expectations or concerns. ‘Reduced Accuracy’ (6 instances) was expressed through statements like “AI might make mis

Increased Assessment Accuracy

doubt its accuracy,” while opinions such as ‘Negative Outlook’ (1 inst

ersonal Information Leak’ (1 instance) were also included.

Positive Outlook

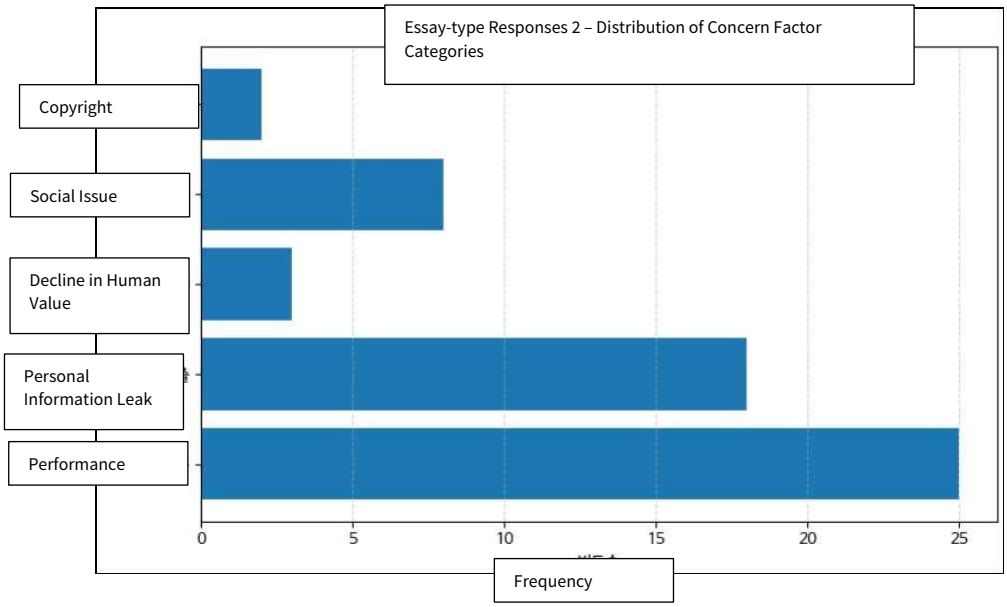


[Figure 4-21] Negative expectation distribution

Of the 78 total responses, 47 were not included in the categories. This is judged to be due to the prevalence of vague or one-sided responses (e.g., “I don't know,” “Don't know,” “None”) rather than specific expectations.

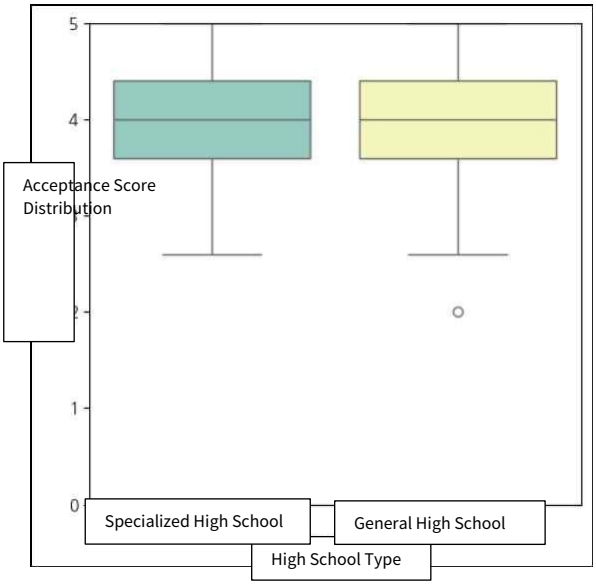
Question 17 (“If you have any concerns or discomfort regarding generative AI-based assessment tools, please feel free to write them down.”) revealed a wider range of concerns. The most frequently cited category was ‘performance-related issues’ (25 instances), with many pointing out the processing accuracy and limitations of generative AI, such as “Interpretation Errors,” “Poor Comprehension,” and “Inability to Handle Complex Questions.”

In addition, ‘‘Personal Information Leaks’ (18 cases), ‘Social Issues’ (8 cases), ‘Copyright Issues’ (2 cases), and ‘Decline in Human Values’ (3 cases) were mentioned.



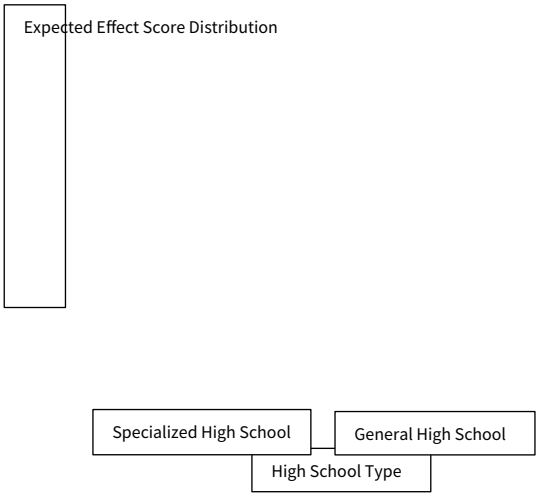
[Figure 4-22] Distribution of concern factor categories

Using Python's scipy and pandas libraries to perform an independent samples t-test, the results of the acceptance analysis between general high school students and specialized high school students showed a t-statistic of 0.1395, a p-value of 0.8892, an average acceptance score of 3.98 for general high schools, and an average acceptance score of 3.97 for specialized high schools. The ANOVA results showed an F-statistic of 0.0193 and a p-value of 0.8896.



[Figure 4-23] Distribution of acceptance for generative AI assessment tools by high school type (Box Plot)

The results of the independent samples t-test conducted in Python for the expected effect analysis between general high school students and specialized high school students showed a t-statistic of 0.0768, a p-value of 0.9375, an average acceptance score of 3.98 for general high schools, and an average acceptance score of 3.97 for specialized high schools. The ANOVA results showed an F-statistic of 0.0061 and a p-value of 0.9378.



[Figure 4-24] Distribution of expected effects of generative AI assessment tools by high school type (Box Plot)

5. Conclusion

5.1 C ysis

The reliability analysis results showed that Cronbach's α value was .737, confirming that the survey's 5-point Likert scale multiple-choice questions achieved good internal consistency (α > 0.7). The mean and standard deviation for each multiple-choice question revealed that questions 1 and 5 had the highest means (M = 4.62, M = 4.64),

indicating that high school students have extensive experience using generative AI in both academic and daily life contexts. However, questions 2 and 14 showed relatively lower average scores (M = 3.29, M = 3.21) compared to other questions, indicating that students recognize the reliability and fairness of generative AI as not high.

By factor, acceptance had the highest mean at 3.9786, followed by concerns (M = 3.7925) and expected effects (M = 3.6918). This indicates that high school students hold a dual recognition of generative AI, showing overall high acceptance while simultaneously harboring both expectations and concerns. Specifically, it shows that they may be slightly more influenced by concerns than by expected effects.

Pearson correlation analysis among acceptance, expected effects, and concerns revealed a significant positive correlation between acceptance and expected effects ($r = .627$, $p < .001$). Conversely, concerns showed no significant correlation with either acceptance or expected effects. That is, students' concerns about generative AI did not appear to significantly influence either acceptance or expected effects.

5.2 Analysis of Recognition of Generative AI-Based Assessment Tools by High School Type

The results of the analysis on the recognition of AI assessment tools by high school type are shown in <Table 5-1> below.

	t-statistic	p-value (Independent samples t-test)	F-statistic	p-value (ANOVA)
Acceptance	0.1395	0.8892	0.0193	0.8896
Expected Effects	0.0768	0.9375	0.0061	0.9378
Concerns	2.8420	0.0051	3.1336	0.0049

[Table 5-1] Statistical values for acceptance, expected effects, and concerns by high school type

The table above suggests the following conclusion.

First, the acceptance analysis results show no statistically significant difference between

general high school students and specialized high school students. Furthermore, analysis of recognition of acceptance revealed that the average score for general high school students was 3.98 points, while that for specialized high school students was 3.97 points, showing almost no difference. Both groups scored close to 4 points, which is considered a ‘positive’ evaluation. This suggests that students generally hold a positive attitude toward AI-based assessment tools.

Second, the results of the expected effect analysis also showed no statistically significant differences. Therefore, both groups expect the AI-based assessment tool to have positive impacts, such as learning diagnosis, preventing cheating, and reducing teachers' workload.

Third, a significant difference existed in concerns between the two groups. The average concern level for general high school students was 3.94, while that for specialized high school students was 3.64, showing a difference of 0.30. That is, the analysis revealed that general high school students expressed a higher level of concern regarding generative AI assessment tools than specialized high school students.

5.3 Expectations and Concerns Regarding AI-Based Assessment Tools

Most students showed a high level of acceptance toward generative AI-based assessment tools. Regarding expected effects, the highest positive responses came from students who believed these tools could reduce teachers' assessment workload and improve the quality of teaching. Next, there was also a high level of recognition that AI technology could bring positive changes to education as a whole in the future. In the open-ended questions, among the positive responses, vague expectations about the future impact on education were the highest. These results show that students recognize that utilizing generative AI as an assessment tool could bring positive changes to the educational environment.

Meanwhile, regarding concerns, the highest level of concern was that AI-based assessment tools might not accurately interpret students' responses, followed by significant concerns about the ambiguous accountability for assessment results. For open-ended questions, concerns were raised about performance issues such as accuracy, hallucination, and the generation of false information. These results show that students recognize the reliability of performance, ethical responsibility in the assessment process, and social repercussions with heightened sensitivity.

5.4 Challenges Encountered during the Survey Process

The following limitations were encountered during the examination process of this research.

First, the survey was conducted at a later time than expected, leaving students with insufficient time. Consequently, it was observed that some students provided fragmented or perfunctory responses to the open-ended questions rather than taking time to carefully consider their answers.

Second, given that the survey participants were high school students, some responses appeared to reflect a lack of seriousness toward the purpose of the questions. For instance, there were cases where respondents wrote only a single word in response to specific questions or provided answers unrelated to the survey questions.

This data was treated as missing values during the preprocessing stage, resulting in significantly fewer open-ended question responses being collected compared to the number of students who responded to the questions.

These factors imposed certain limitations on the interpretation of results. Therefore, subsequent studies should consider providing sufficient explanations and sample answers for each question, as well as planning survey schedules aligned with the academic calendar.

5.5 Limitations and Suggestions for Future Research

This research examined high school students' recognition of AI-based assessment tools by categorizing it into acceptance, expected effects, and concerns. While it serves as a foundation for introduction, it has several limitations. Therefore, the following research directions are proposed for subsequent studies.

First, since education policy must reflect the recognition of parents, educational administrators, etc., it is necessary to conduct recognition studies targeting diverse educational stakeholders, including those groups.

Second, the majors of the specialized high school students in this research are concentrated in IT-related fields such as software and IT security. Therefore, it is difficult to consider them representative of the entire population of specialized high school students across diverse fields like arts, physical education, and business. Consequently, it is necessary to additionally consider differences in recognition based on major field.

Third, the sample size was limited to 159 participants, making it difficult to generalize the findings. Therefore, additional research with a sample size of 300 or more is recommended.

Through such follow-up research, we anticipate gaining a more concrete understanding of the acceptance potential, expected effects, and level of concern when introducing AI-based assessment tools into educational settings. This will serve as the foundation for effective policy proposals and introduction strategies.

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